

VARUN GUNDA

Phone: (562) 294-0602 | Email: gunda.varun.05@gmail.com | [LinkedIn](#) | Location: USA

SUMMARY

- Machine Learning Engineer with 5+ years of experience designing, developing, and deploying AI and data-driven solutions across healthcare and enterprise domains.
- Skilled in building end-to-end ML pipelines, NLP systems, and predictive models using Python, TensorFlow, PyTorch, Scikit-learn, and XGBoost.
- Experienced in MLOps and cloud integration, leveraging AWS SageMaker, Vertex AI, Airflow, Docker, and MLflow for scalable, production-ready model deployment.
- Strong background in data engineering, automation, and model explainability using PySpark, BigQuery, and SHAP, ensuring accuracy, compliance, and transparency.
- Proven ability to collaborate across multidisciplinary teams to deliver impactful, high-performance AI solutions that drive operational efficiency and business value.

SKILLS

- **Programming & Scripting:** Python, SQL, Scala, Java, Bash
- **Machine Learning & Deep Learning:** TensorFlow, PyTorch, Scikit-learn, XGBoost, Hugging Face, OpenCV
- **Modeling & Algorithms:** Deep Learning (CNNs, RNNs, Transformers), Time Series Forecasting, NLP, Recommendation Systems, Predictive Analytics
- **Data Engineering & Processing:** Pandas, NumPy, PySpark, Apache Airflow, Hive, BigQuery
- **MLOps & Deployment:** MLflow, AWS SageMaker, Docker, Kubernetes, REST APIs, CI/CD Pipelines
- **Cloud Platforms:** AWS (S3, EC2, Lambda, SageMaker), GCP (BigQuery, AI Platform)
- **Experimentation & Monitoring:** A/B Testing, Model Monitoring, Explainability (SHAP, LIME), TensorBoard
- **Collaboration & Workflow:** Git, Jira, Confluence, Agile/Scrum

WORK EXPERIENCE

Humana

Feb 2025 – Present

Machine Learning Engineer

- Spearheading the development of Ambient Clinical Assistant, an AI-driven system that captures and summarizes physician-patient dialogues to automate EHR documentation using advanced speech and language models.
- Engineered and optimized transformer architectures (Whisper, Flan-T5) to enhance contextual understanding and medical entity extraction accuracy.
- Architected a real-time ML pipeline leveraging Vertex AI, Airflow, and MLflow for scalable deployment, experiment tracking, and model lifecycle management.
- Established robust explain ability and compliance mechanisms through SHAP-based interpretation tools and governance dashboards to meet healthcare regulatory standards.
- Partnered with clinicians and cross-functional teams to validate pilot deployments, achieving a 40% reduction in manual documentation time and improving workflow efficiency.

Humana

Jul 2024 – Dec 2024

Machine Learning Engineer - Intern

- Contributed to the Claims Cost Prediction Model, developing data-driven solutions to forecast healthcare claim expenses and support risk-adjusted pricing strategies.
- Processed and engineered large-scale claims data using PySpark and Hive, improving data quality and pipeline efficiency for model training.
- Built and optimized regression and ensemble models (XGBoost, LightGBM) to accurately estimate claim cost trajectories across diverse member populations.
- Deployed predictive models on AWS SageMaker, integrating performance monitoring and automated retraining workflows for sustained model accuracy.
- Collaborated with actuarial and data analytics teams to deliver insights via Tableau dashboards, increasing forecast precision by 15% compared to legacy models.

Dell Technologies

Jul 2018 – Dec 2022

Machine Learning Engineer

- Directed the Predictive Maintenance for Server Hardware initiative, implementing advanced ML systems to anticipate server component failures and ensure proactive maintenance across enterprise environments.
- Developed robust time-series forecasting and anomaly detection models using large-scale telemetry and performance datasets to predict hardware degradation trends.
- Constructed high-throughput data processing pipelines with Apache Spark, Kafka, and SQL, enabling real-time analytics and continuous model updates.
- Utilized TensorFlow and Scikit-learn frameworks to design and optimize predictive algorithms, resulting in a 25% decrease in unplanned downtime.
- Automated deployment, testing, and retraining cycles using Jenkins CI/CD, ensuring consistent performance and rapid integration of new model versions.
- Embedded predictive outputs into Dell ProSupport dashboards, empowering engineers with real-time visualizations through Grafana and API integrations.

- Collaborated closely with reliability and infrastructure teams to integrate predictive insights into maintenance workflows, reducing mean time to resolution (MTTR) by 30%.
- Performed detailed model evaluation and feature attribution studies to enhance interpretability, improve precision, and strengthen trust in AI-driven maintenance predictions.

PROJECTS

Retail Sales Forecasting System

- Developed time-series forecasting models using Prophet and LSTM networks to predict multi-store sales trends.
- Conducted feature engineering and validation using Pandas, Scikit-learn, and NumPy on retail datasets from Kaggle.
- Designed forecasting pipelines to support demand planning and sales trend visualization for business insights.

Credit Risk Assessment Model

- Built a classification pipeline using Logistic Regression, XGBoost, and Random Forest to predict loan defaults.
- Applied SMOTE for class imbalance handling and GridSearchCV for hyperparameter tuning and model optimization.
- Utilized SHAP-based interpretability techniques to explain key financial risk factors influencing model predictions.

EDUCATION

Master in Computer Science

Jan 2023 – Dec 2024

California State University, Long Beach – Long Beach, CA | GPA: 3.3